//Example 1

```
abstract class Shape {
   public abstract void draw();
}

 class Circle extends Shape {
   public void draw() {
      System.out.println("Circle");
   }
}

class Rectangle extends Shape {
   public void draw() {
      System.out.println("Rectangle");
   }
}

class Triangle extends Shape {
   public void draw() {
      System.out.println("Triangle");
   }
}

class AbstractDemo
{
   public static void main(String args[])
   {

      Shape shape = new Circle();
      shape.draw();
      shape = new Rectangle();
      shape.draw();
      shape = new Triangle();
      shape.draw();

   }
}
```

Example2

```
abstract class Shape
{
  public abstract void draw();
  public abstract void paint();
}
```

```
class Circle extends Shape
{
  public  void draw()
  {
                    System.out.println("drawing Circle");
  }

  public  void paint()
  {
                    System.out.println("painting Circle");
  }
}


class AbstractDemo
{
 public static void main(String args[])
 {
                    Shape shape = new Circle();
                    shape.draw();
                    shape.paint();
 }
}
```

//Example3

```
abstract class Shape
{
 public abstract void draw();
 public abstract void paint();
}

abstract class Circle extends Shape
{
 public  void draw()
 {
  System.out.println("Circle");
 }
}

class Rectangle extends Shape
{
 public  void draw()
 {
  System.out.println("drawing Circle");
```

```
 }
 public  void paint()
 {
  System.out.println("painting Circle");
 }

}

class AbstractDemo
{
 public static void main(String args[])
 {
          Shape rectangle = new Rectangle(); //allowed
          rectangle.draw();
  rectangle.paint();
  // Circle circle = new Circle();
  //Rectangle rectangle = Rectangle();
  //Shape circle = new Circle(); //?
 }
}
```

Example 4

```
abstract class Shape
{
  void draw()
  {
            System.out.println("hi");
  }
}

class AbstractDemo
{
 public static void main(String args[])
 {
  Shape shape = new Shape();
 }
}
```

//Example 5

//Creating object for abstract class

```
   abstract class Shape
   {
    abstract void draw();
   }
```

```
class AbstractDemo
{
  public static void main(String args[])
  {
    Shape shape = new Shape();
    shape.draw();
  }
}
```

Interface

Example 1

```
interface Shape
{
        void draw();
}

//Implementation class of Shape
class Circle implements Shape
{
  public void draw()
  {
                System.out.println("Draws a circle");
  }
}

class InterfaceDemo
{
        public static void main(String args[])
        {
                Shape shape  = new Circle();

                shape.draw();
        }
}
```

Example 2

```
interface Shape
{
 void draw();
 void paint();
}


class Circle  implements Shape
{
```

```
  public void draw(){
                System.out.println("draws a circle");
  }
}

class InterfaceDemo
{
  public static void main(String args[])
  {
          Shape shape  = new Circle();
          shape.draw();
  }
}
```

Default and static method

```
interface SampleI {

    public void m1();
    // public void m2();
    // public void m4();

    default void m3DefaultMethod() {
        System.out.println("This is default method");
    }

    default void m2() {
        System.out.println("This is default method");
    }


      default void m3() {
        System.out.println("This is default method");
    }

    static void s1() {
        System.out.println("static method");
    }

    default void m9() {
        System.out.println("This is default method");
    }
}
```

```
class A1 implements SampleI {

    public void m1() {
        System.out.println("Method 1 implementation of
class A");
    }

    public void m2() {
        System.out.println("Method 2 implementation of
class A");
    }

}

class B1 implements SampleI {

    public void m1() {
        System.out.println("Method 1 implementation of
class B");
    }

    public void m2() {
        System.out.println("Method 2 implementation of
class B");
    }

   public void  m3DefaultMethod()
   {
       System.out.println("Overriding default method");
   }

}


class C1 implements SampleI {

    public void m1() {
        System.out.println("Method 1 implementation of
class C");
    }
```

```
    public void m2() {
        System.out.println("Method 2 implementation of
class C");
    }
}

class D implements SampleI {

    public void m1() {
        System.out.println("Method 1 implementation of
class D");
    }

    public void m2() {
        System.out.println("Method 2 implementation of
class D");
    }
}

public class DefaultAndStaticMethod {

    public static void main(String[] args) {
        SampleI obj1 = new A1();
        obj1.m1();
        obj1.m2();

        SampleI.s1();

        SampleI obj2 = new B1();
        obj2.m1();
        obj2.m2();
        obj2.m3DefaultMethod(); //accessing default method

        SampleI obj3 = new B1();
        obj3.m1();
        obj3.m2();

        SampleI.s1(); //accessing static method

        B1 b1 = new B1();
        b1.m3DefaultMethod();
```

```
    }
}
```

Note:

In Java, a static method defined in an interface belongs only to that interface and is not inherited by the implementing class.

To access the static method of the interface we have to use interfacename.static method name